



Python Programming - 2301CS404

Lab - 9

Roll No : 310

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**01) Write a function to calculate BMI given mass and height.
(BMI = mass/h**2)**

```
In [4]: mass = float(input("Enter the mass(in kg):"))  
h = float(input("Enter the height(in m):"))  
def calBMI(mass,h) :  
    return mass/h**2  
print(calBMI(mass,h))
```

24.489795918367346

02) Write a function that add first n numbers.

```
In [9]: n = int(input("Enter the number:"))  
def add(n):  
    for i in range(1,n+1) :  
        return n*(n+1)/2  
print(int(add(n)))
```

210

03) Write a function that returns 1 if the given number is Prime or 0 otherwise.

```
In [13]: n = int(input("Enter the number:"))  
def isPrime(n) :  
    count=0  
    for i in range(1,n+1) :  
        if(n%i==0) :  
            count+=1  
    if(count==2) :
```

```

        return 1
    else :
        return 0
print(isPrime(n))

```

1

04) Write a function that returns the list of Prime numbers between given two numbers.

```

In [14]: start = int(input("Enter the number-1:"))
end = int(input("Enter the number-2:"))
def get_primes(start, end):
    prime_list = []
    for num in range(start, end + 1):
        if num > 1:
            for i in range(2, int(num**0.5) + 1):
                if (num % i) == 0:
                    break
            else:
                prime_list.append(num)
    return prime_list
print(get_primes(start,end))

```

[11, 13, 17, 19]

05) Write a function that returns True if the given string is Palindrome or False otherwise.

```

In [17]: s1 = input("Enter the string:")
rev = lambda s1: s1[::-1]
reverse_s1 = rev(s1)
if(reverse_s1==s1) :
    print("True")
else :
    print("False")

```

True

06) Write a function that returns the sum of all the elements of the list.

```

In [22]: n = int(input("Enter the size of the list:"))
l1 = list(map(int,input("Enter the elements:").split()))
def sumOfList(l1) :
    return sum(l1)
print(sumOfList(l1))

```

150

07) Write a function to calculate the sum of the first element of each tuples inside the list.

```

In [28]: n = int(input("Enter the size of tuple:"))
l1 = []
for i in range(n) :
    t1 = tuple(map(int,input("Enter the elements of tuple").split()))

```

```

l1.append(t1)
def sumOfFirstOfTuple(t_list) :
    total = 0
    for i in range(n) :
        total+=l1[i][0]
    return total
print(sumOfFirstOfTuple(l1))

```

5

08) Write a recursive function to find nth term of Fibonacci Series.

```

In [29]: n = int(input("Enter the number:"))
def fibbo(n) :
    if n==0 :
        return 0
    elif n==1 :
        return 1
    else :
        return fibbo(n-1) + fibbo(n-2)
print(fibbo(n-1))

```

3

09) Write a function to get the name of the student based on the given rollno.

Example: Given dict1 = {101:'Ajay', 102:'Rahul', 103:'Jay', 104:'Pooja'} find name of student whose rollno = 103

```

In [30]: n = int(input("Enter the no. of students:"))
d1 = {}
for i in range(n) :
    roll = int(input("Enter the rollNo of student:"))
    name = input("Enter the name of student:")
    d1[roll] = name
roll_key = int(input("Enter roll to find the name of student:"))
def getStudentName(d1,roll) :
    if roll in d1 :
        return d1[roll]
    else :
        return "Roll No not found"
print(getStudentName(d1,roll_key))

```

Jay

10) Write a function to get the sum of the scores ending with zero.

Example : scores = [200, 456, 300, 100, 234, 678]

Ans = 200 + 300 + 100 = 600

```

In [31]: n = int(input("Enter the no. of scores:"))
l1 = list(map(int,input("Enter the scores:").split()))
def sumOfScore(l1) :

```

```

total = 0
for i in l1 :
    if i%10==0 :
        total+=i
return total
print(sumOfScore(l1))

```

600

11) Write a function to invert a given Dictionary.

hint: keys to values & values to keys

Before : {'a': 10, 'b':20, 'c':30, 'd':40}

After : {10:'a', 20:'b', 30:'c', 40:'d'}

```

In [33]: d1 = {}
n = int(input("Enter the size of the dictionary: "))
for i in range(n):
    k = input("Enter the key:")
    v = int(input("Enter the value:"))
    d1[k] = v
inv_di = {}
for k,v in d1.items():
    inv_di[v] = k
print(inv_di)

```

{10: 'a', 20: 'b', 30: 'c', 40: 'd'}

12) Write a function that returns the number of uppercase and lowercase letters in the given string.

example : Input : s1 = AbcDEfgh ,Ouptput : no_upper = 3, no_lower = 5

```

In [36]: s1 = input("Enter the string:")
def noOfUpperLower(s1) :
    upper = 0;lower = 0
    for i in s1 :
        if i.isupper() :
            upper +=1
        else :
            lower +=1
    return upper,lower
upper,lower = noOfUpperLower(s1)
print(upper)
print(lower)

```

3

5

13) Write a lambda function to get smallest number from the given two numbers.

```

In [37]: a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

```

```
smallest = lambda a, b: a if a < b else b
print(smallest(a,b))
```

2

14) For the given list of names of students, extract the names having more that 7 characters. Use filter().

```
In [4]: n = int(input("Enter the no. of students:"))
l1 = list(map(str,input("Enter the students:").split()))
def getNames(name) :
    return len(name) > 7
print(list(filter(getNames,l1)))
```

```
['Alexander', 'Christopher']
```

15) For the given list of names of students, convert the first letter of all the names into uppercase. use map().

```
In [5]: n = int(input("Enter the no. of students:"))
l1 = list(map(str,input("Enter the students:").split()))
def firstCapital(name) :
    return name[0].upper() + name[1:] if name else name
print(list(map(firstCapital,l1)))
```

```
['Amit', 'Neha', 'Ravi']
```

16) Write udfs to call the functions with following types of arguments:

1. Positional Arguments
2. Keyword Arguments
3. Default Arguments
4. Variable Length Positional(*args) & variable length Keyword Arguments (**kwargs)
5. Keyword-Only & Positional Only Arguments

```
In [13]: # 1. Positional Arguments
def add(a, b):
    print("Sum=", a + b)
add(10, 20)
# 2. Keyword Arguments
def add(a,b) :
    print("Sum=", a + b)
add(b=20,a=10)
# 3. Default Arguments
def greet(name, msg="Good Morning"):
    print(msg, name,end=",")
greet("Amit")
greet("Ravi", "Good Evening")
# 4. Variable Length Positional Arguments (*args) & variable Length Keyword Argu
def total(*numbers):
    s = 0
    for i in numbers:
        s += i
    print("\nTotal:", s)
total(10, 20, 30, 40)
```

```
def show_details(**info) :  
    for key, value in info.items() :  
        print(key, ":", value,end=",")  
show_details(name="Ajay", age=22, city="Rajkot")  
# 5. Keyword-Only Arguments & Positional Only Arguments  
def divide(a, b, /):  
    print("\nResult:", a / b)  
divide(10, 2)  
def login(*, username, password):  
    print("Username:", username,end=",")  
    print("Password:", password)  
login(username="admin", password="1234")
```

Sum= 30

Sum= 30

Good Morning Amit,Good Evening Ravi,

Total: 100

name : Ajay,age : 22,city : Rajkot,

Result: 5.0

Username: admin>Password: 1234